Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 4 — Trigonometric addition formulae Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第4章 · 三角加法公式
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C04-R01 · P3 · Chapter 4

Angles α, β are acute, with $\sin\alpha = 3/5$ and $\tan\beta = 5/12$.

(a) Find $\cos\alpha$ and $\sin\beta$ exactly. (2)

(b) Find $\sin(\alpha+\beta)$ and $\cos(\alpha-\beta)$ exactly. (3)

(Total for Question 1 is 5 marks)

$F(x) = 8\sin x + 15\cos x$.

(a) Express F as $R\sin(x + \alpha)$, where $R > 0$ and $0 < \alpha < \pi/2$. (3)

(b) Define $G(x) = 15\sin x - 8\cos x$.

(i) show that $F(x)^2 + G(x)^2 = 289$, (3)

(ii) find all possible values of $G(x)$ when $F(x) = 8$. (3)

(Total for Question 2 is 9 marks)

Without calculator technology,

(a) find the exact value of $\sin 75^\circ$, (3)

(b) find the exact value of $\tan 15^\circ$, with a rational denominator. (3)

(Total for Question 3 is 6 marks)

For $0 \leq x < 2\pi$, $\sin(2x) = \cos x$.

(a) Solve the equation exactly. (4)

(b) Find the solutions lost by dividing both sides by $\cos x$. (2)

(Total for Question 4 is 6 marks)

On their common domain,

(a) prove that $\dfrac{1-\cos 2x}{\sin 2x} = \tan x$. (3)

(b) find the exact value of this expression when $x = \pi/12$. (3)

(Total for Question 5 is 6 marks)

$\tan\alpha = 2$ and $\tan\beta = 1/3$, with both angles acute.

(a) Find $\tan(\alpha+\beta)$ exactly and determine whether $\alpha+\beta$ is acute or obtuse. (4)

(b) Find $\tan(\alpha-\beta)$ exactly. (2)

(Total for Question 6 is 6 marks)

$H(x) = 5\cos x - 12\sin x$.

(a) Write $H = R\cos(x + \alpha)$ for $R > 0$, $0 < \alpha < \pi/2$. (3)

(b) Find the total length of the intervals in $0 \le x < 2\pi$ where $H(x) > 0$. (3)

(Total for Question 7 is 6 marks)

8.

For $0 \leq x \leq \pi$, $\cos 2x + \cos x = 0$.

(a) Find all exact solutions. (4)

(b) Find $\sin 2x$ at each solution. (2)

(Total for Question 8 is 6 marks)

9. P3-C04-R09 · P3 · Chapter 4

For $0 < \theta < \pi$, $\cos\theta = -3/5$.

(a) Find $\sin(\theta/2)$ and $\cos(\theta/2)$ exactly. (3)

(b) Find $\tan(\theta/2)$ exactly. (2)

(Total for Question 9 is 5 marks)

A signal is modelled by $S(t) = 4 + 3\cos t + 3\sqrt{3}\sin t$, where t is in radians.

(a) Write S as $4 + R\cos(t - \alpha)$, with $R > 0$ and $0 < \alpha < \pi/2$. (3)

(b) On $0 \leq t < 2\pi$, find

(i) the maximum signal and the time it first occurs, (2)

(ii) all times for which the signal is 4. (3)

(Total for Question 10 is 8 marks)

Prove the identities.

(a) $\sin(x+\alpha)+\sin(x-\alpha)=2\sin x\cos\alpha.$ (3)

(b) $\cos(x-\alpha)-\cos(x+\alpha)=2\sin x\sin\alpha.$ (3)

(Total for Question 11 is 6 marks)

For $0 \leq x < 2\pi$, $\sin(x + \pi/6) = \sin(x - \pi/6)$.

(a) Use addition formulae to simplify the equation. (3)

(b) Hence solve exactly. (2)

(Total for Question 12 is 5 marks)

$\cos\theta = 5/13$ and $3\pi/2 < \theta < 2\pi$.

(a) Find $\sin 2\theta$ and $\cos 2\theta$ exactly. (4)

(b) Find $\tan 2\theta$ exactly. (2)

(Total for Question 13 is 6 marks)

For $0 < x < \pi$, $\tan 2x = \tan x$.

(a) Find all exact solutions, checking where the original tangent functions are defined. (3)

(b) Find all solutions if the domain is instead $0 \leq x \leq \pi$. (2)

(Total for Question 14 is 5 marks)

$F(x) = \cos^2 x - \sin^2 x + 2\sin x \cos x$.

(a) Express F as $R\cos(2x - \alpha)$ with exact R, α. (3)

(b) Find the least positive x at which F reaches its maximum. (3)

(Total for Question 15 is 6 marks)

A triangle has sides $AB = AC = 10$ cm and included angle 2θ, where $0 < \theta < \pi/2$.

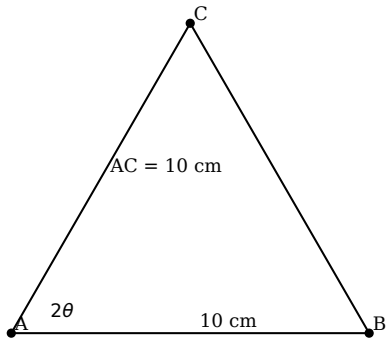


Figure 16

(a) Show that its area is $100 \sin\theta \cos\theta$. (3)

(b) Find all values of θ for which the area is $25\sqrt{3}$ cm^2. (3)

(Total for Question 16 is 6 marks)

For $0 \leq x < 2\pi$, $\sin(x + \pi/3) + \sin(x - \pi/3) = \cos 2x$.

(a) Use addition formulae to show that the left-hand side is $\sin x$. (3)

(b) Hence find every exact solution of the equation. (4)

(Total for Question 17 is 7 marks)

Let $\alpha = \arctan(1/2)$ and $\beta = \arctan(1/3)$.

(a) Find $\tan(\alpha + \beta)$ and use the principal ranges to determine $\alpha + \beta$ exactly. (4)

(b) Find $\tan(\alpha - \beta)$ and hence give the exact value of $\tan(2\alpha - \pi/4)$. (3)

(Total for Question 18 is 7 marks)

19. P3-C04-R19 · P3 · Chapter 4

For $0 \leq x < \pi$, $\cos(2x - \pi/4) = 0$.

(a) Find all exact solutions. (4)

(b) Sketch one period of $y = \cos(2x - \pi/4)$ on the stated interval, labelling the zeros and its maximum point. (3)

(Total for Question 19 is 7 marks)

Prove that $\tan(x + \pi/4) = \dfrac{1 + \tan x}{1 - \tan x}$ on the common domain.

(a) Derive the formula from the tangent addition formula. (3)

(b) Solve $\tan(x + \pi/4) = 3$ for $0 < x < \pi/2$. (3)

(Total for Question 20 is 6 marks)

$P(x) = 9\sin x + 12\cos x$.

(a) Find the greatest possible value of P and write an equation for the angles attaining it. (4)

(b) A modified signal is $Q(x) = P(x) - P(x + \pi)$. Find its maximum without re-expanding a new auxiliary angle. (3)

(Total for Question 21 is 7 marks)

For $0 \leq x \leq \pi$, $\sin(2x) = \sqrt{3}\sin x$.

(a) Find all exact solutions. (4)

(b) Find the unique interior solution and the exact value of $\cos 2x$ there. (3)

(Total for Question 22 is 7 marks)

For $0 < \theta < \pi$, $\sin\theta = 7/25$ and θ is obtuse.

(a) Find $\tan(\theta/2)$ exactly. (3)

(b) Find $\cos^2(\theta/2)$ exactly. (2)

(Total for Question 23 is 5 marks)

The identity $\sin 3x = 3\sin x - 4\sin^3 x$ is to be established.

(a) Use $\sin(2x + x)$ and double-angle formulae to prove it. (4)

(b) Hence solve $\sin 3x = \sin x$ for $0 \le x \le \pi/2$. (3)

(Total for Question 24 is 7 marks)

For $0 \leq x < 2\pi$, $\cos(x + \pi/3) = \cos(x - \pi/3)$.

(a) Simplify using addition formulae and solve. (4)

(b) Find the common cosine value at each solution. (2)

(Total for Question 25 is 6 marks)

A rotating point has height $h = 10 + 6\sin t - 8\cos t$ metres, where $0 \leq t < 2\pi$.

(a) Express h as $10 + R\sin(t - \alpha)$ for $R > 0$ and acute α. (3)

(b) Find

(i) the minimum height and the time at which it occurs, (3)

(ii) the total length of the time intervals where $h > 10$. (2)

(Total for Question 26 is 8 marks)

Prove that $\cos^4 x - \sin^4 x = \cos 2x$.

(a) Give an algebraic proof using a difference of squares. (3)

(b) Hence solve $\cos^4 x - \sin^4 x = -1/2$ for $0 \le x < \pi$. (3)

(Total for Question 27 is 6 marks)

A line segment of length 20 cm rotates through an acute angle θ. Its horizontal and vertical projections have sum 28 cm.

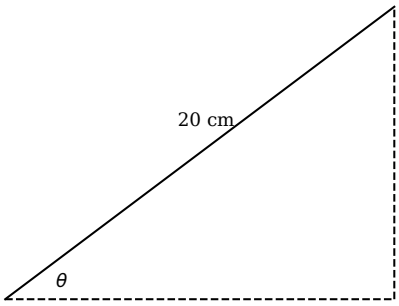


Figure 28

(a) Show that $\sin\theta + \cos\theta = 7/5$ and find $\sin 2\theta$. (3)

(b) Find the two possible angles to 1 decimal place. (3)

(Total for Question 28 is 6 marks)

Figure 29 shows the curve $y = 6\sin x + 8\cos x$ for $0 \leq x \leq 2\pi$.

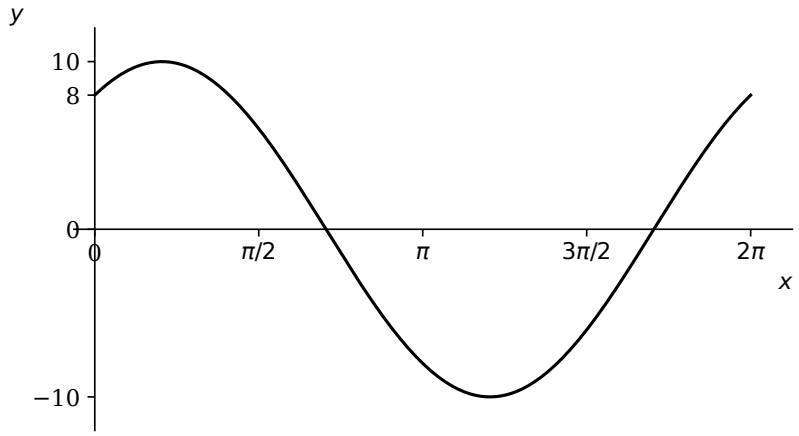


Figure 29

(a) Write $6\sin x + 8\cos x$ as $R\cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \pi/2$. Give α to 4 decimal places. (3)

(b) The horizontal line $y = k$ meets the curve at two distinct points in $0 \leq x < 2\pi$.

(i) Find the full range of k. (2)

(ii) For $k = 5$, find both x-coordinates to 3 decimal places. (2)

(Total for Question 29 is 7 marks)

Two rods of lengths 4 m and 3 m meet at a hinge. They make angles θ and $(\pi/2 - \theta)$, respectively, with the horizontal, and both extend to the right and upwards from the hinge. Here $0 < \theta < \pi/2$. The sum of their vertical projections is H m.

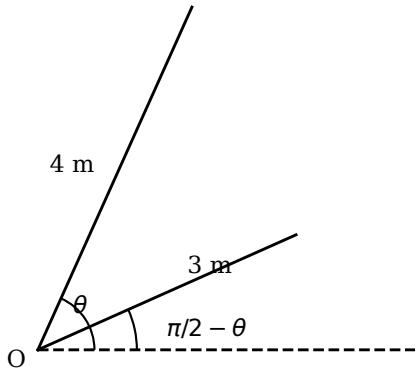


Figure 30

(a) Show that $H = 4\sin\theta + 3\cos\theta$ and express it in the form $5\sin(\theta + \beta)$, specifying β. (3)

(b) Find

(i) the greatest value of H and the exact value of θ at which it occurs. (2)

(ii) all possible values of θ for which $H = 9/2$, to 3 decimal places. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C04-R01 · 5 marks

(a) · 2 marks

$4/5$ and $5/13$.

两参考直角三角形分别3、4、5及5、12、13，锐角取正。

M1	两参考三角形
A1	两值

(b) · 3 marks

$56/65$ and $63/65$.

sin和=(3/5)(12/13)+(4/5)(5/13)=56/65；cos差=(4/5)(12/13)+(3/5)(5/13)=63/65。

M1	正弦和公式
M1	余弦差公式
A1	两精确值

Question 2

P3-C04-R02 · 9 marks

(a) · 3 marks

$R = 17$, $\alpha = \arctan(15/8)$.

比较Rcosα=8、Rsinα=15，平方和R17。

M1	正确两系数对应
M1	R
A1	精确alpha

(b) Define $G(x) = 15\sin x - 8\cos x$.

(i) · 3 marks

Cross terms cancel; sum is $289(\sin^2 x + \cos^2 x) = 289$.

展开两平方时交叉项为±240sinxcosx，抵消。

M1	展开并抵消交叉项
M1	平方和恒等式
A1	结论

(ii) · 3 marks

$G(x) = \pm 15$.

由(i)G^2=289−64=225，两符号在对应相位均能实现。

M1	代入平方和
M1	两符号
A1	值

Question 3

P3-C04-R03 · 6 marks

(a) · 3 marks

$(\sqrt{6}+\sqrt{2})/4$.

以45°+30°展开正弦和公式。

M1	选择标准角和
M1	精确三角值
A1	最简根式

(b) · 3 marks

$2-\sqrt{3}$.

用45°−30°正切差公式，再有理化(1−1/√3)/(1+1/√3)。

M1	正切差公式
M1	有理化
A1	精确值

Question 4

P3-C04-R04 · 6 marks

(a) · 4 marks

$x = \pi/6, \pi/2, 5\pi/6, 3\pi/2.$

用2sinxcosx=cosx，分解cosx(2sinx−1)=0，列两分支。

M1	双角代换
M1	零因子保留
M1	全部角
A1	四解

(b) · 2 marks

$\pi/2, 3\pi/2.$

原式在cos零点也满足，直接除法会丢掉这两根。

M1	检查零因子
A1	两角

Question 5

P3-C04-R05 · 6 marks

(a) · 3 marks

Use $1-\cos 2x = 2\sin^2 x$ and $\sin 2x = 2\sin x\cos x$.

原分母不为0保证sin与cos都非零，约分后sin/cos。

M1	两双角恒等式
M1	合法约分
A1	目标式

(b) · 3 marks

$2-\sqrt{3}$.

引用(a)变为tan15°，由角差公式得精确值。

M1	使用(a)
M1	标准角差
A1	值

Question 6

P3-C04-R06 · 6 marks

(a) · 4 marks

7; acute.

正切和(2+1/3)/(1−2/3)=7；两锐角之和在(0,π)，正切正所以仍锐角。

M1	正切和公式
M1	精确比值
M1	角和范围
A1	类型

(b) · 2 marks

1.

(2−1/3)/(1+2/3)=1。

M1	正切差公式
A1	精确值

Question 7

P3-C04-R07 · 6 marks

(a) · 3 marks

$R = 13,\ \alpha = \arctan(12/5).$

比较Rcosα=5、Rsinα=12，注意余弦和公式的负号。

M1	系数比较
M1	R
A1	alpha

(b) · 3 marks

$\pi.$

辅助角仅水平移相，完整一周期内cos正的输入总长度π；严格端点不改变总长度。

M1	化为正余弦区间
M1	完整周期长度
A1	结果

Question 8

P3-C04-R08 · 6 marks

(a) · 4 marks

$x = \pi/3, \pi$.

cos2x=2c²−1，得2c²+c−1=(2c−1)(c+1)，按0至π取每种一角。

M1	双角余弦代换
M1	分解
M1	域筛选
A1	两角

(b) · 2 marks

$\sqrt{3}/2$ and 0, respectively.

对应输入2π/3和2π。

M1	正确倍角输入
A1	两值

Question 9

P3-C04-R09 · 5 marks

(a) · 3 marks

$2/\sqrt{5}$ and $1/\sqrt{5}$.

半角在第一象限，两正平方根：$\sin^2$半角$=(1-\cos\theta)/2=4/5$，$\cos^2$半角$=1/5$。

M1	两半角平方关系
M1	正分支
A1	两值

(b) · 2 marks

2.

两半角比相除，公共根式抵消。

M1	半角比商
A1	值

Question 10

P3-C04-R10 · 8 marks

(a) · 3 marks

$S = 4 + 6\cos(t - \pi/3)$.

比较6cosα=3、6sinα=3√3得απ/3。

M1	正确系数对应
M1	R
A1	精确相位

(b) On $0 \le t < 2\pi$, find

(i) · 2 marks

Maximum 10 at $t = \pi/3$.

cos输入0时取1，检查相位落域内。

M1	余弦最大
A1	数值与位置

(ii) · 3 marks

$t = 5\pi/6, 11\pi/6$.

cos(t−π/3)=0，输入π/2+kπ，筛选两t。

M1	零余弦输入
M1	相位还原
A1	两t

Question 11

P3-C04-R11 · 6 marks

(a) · 3 marks

Expand both sine terms; the $\cos x \sin \alpha$ terms cancel.

逐项用正弦和差公式，保留sinxcosα两份。

M1	两正确展开
M1	抵消交叉项
A1	目标式

(b) · 3 marks

Expand both cosine terms; the $\cos x \cos \alpha$ terms cancel.

减去第二式时改变其两项符号，最终正的两倍sinxsinα。

M1	两余弦展开
M1	正确相减符号
A1	恒等式

Question 12

P3-C04-R12 · 5 marks

(a) · 3 marks

$\cos x = 0$.

两边展开后sinxcosπ/6抵消，余2cosxsinπ/6=0。

M1	两和差展开
M1	正确抵消
A1	简化条件

(b) · 2 marks

$x = \pi/2, 3\pi/2$.

半周两余弦零点，均在指定域内。

M1	正确零点筛选
A1	两根

Question 13

P3-C04-R13 · 6 marks

(a) · 4 marks

$-120/169$ and $-119/169$.

第四象限$\sin\theta=-12/13$；双角$\sin 2=2\sin\cos$，$\cos 2=2\cos^2-1$。

M1	象限正弦
M1	正弦双角
M1	余弦双角
A1	两值

(b) · 2 marks

$120/119$.

两双角比相除，负除负得正。

M1	比值商
A1	精确值

Question 14

P3-C04-R14 · 5 marks

(a) · 3 marks

No solutions.

同正切意味着2x=x+kπ，所以x=kπ，开区间没有；或双角公式t需检其分母且不丢cosx=0的原无定义点。

M1	正切周期相等条件
M1	原定义及开区间筛选
A1	无解

(b) · 2 marks

$x = 0, \pi.$

两个边界正切和双倍正切都定义且都为0。

M1	检查新增端点
A1	两根

Question 15

P3-C04-R15 · 6 marks

(a) · 3 marks

$F = \sqrt{2}\cos(2x - \pi/4)$.

前两项cos2x，末项sin2x，两系数1得R√2、相位π/4。

M1	双角化简
M1	辅助角系数
A1	形式

(b) · 3 marks

$x = \pi/8$.

2x−π/4=2kπ，最小正解π/8，最大值√2。

M1	最大余弦输入条件
M1	最小正支
A1	x

Question 16

P3-C04-R16 · 6 marks

(a) · 3 marks

Area $50\sin 2\theta = 100\sin\theta\cos\theta$.

两边夹角面积式，再用双角公式。

M1	正确面积50sin2theta
M1	双角转换
A1	目标式

(b) · 3 marks

$\theta = \pi/6, \pi/3$.

sin2θ=√3/2，输入2θ∈(0,π)，两角π/3、2π/3再除2。

M1	正确面积方程
M1	输入范围与两解
A1	两theta

Question 17

P3-C04-R17 · 7 marks

(a) · 3 marks

$2\sin x\cos(\pi/3)=\sin x$.

正弦和差展开后交叉项相消，余弦π/3为1/2。

M1	两加角展开
M1	抵消交叉项
A1	简式

(b) · 4 marks

$x=\pi/6,5\pi/6,3\pi/2$.

令s=sinx，用cos2x=1−2s²得(2s−1)(s+1)=0；两正弦值产生三角。

M1	双角转二次式
M1	因式分解
M1	全区间筛选
A1	三解

Question 18

P3-C04-R18 · 7 marks

(a) · 4 marks

$\tan(\alpha+\beta)=1, \alpha+\beta=\pi/4.$

和公式给(1/2+1/3)/(1−1/6)=1；两主值均锐、和在(0,π)，正切正选择第一象限π/4。

M1	正切和公式
M1	精确比1
M1	合角范围判断
A1	精确合角

(b) · 3 marks

Both equal $1/7$.

差公式(1/2−1/3)/(1+1/6)=1/7；由(a)有β=π/4−α，所以α−β=2α−π/4。

M1	正切差公式
M1	用合角关系重写
A1	两比值

Question 19

P3-C04-R19 · 7 marks

(a) · 4 marks

$x = 3\pi/8, 7\pi/8$.

输入从−π/4到7π/4，余弦零点π/2、3π/2；加π/4除2。

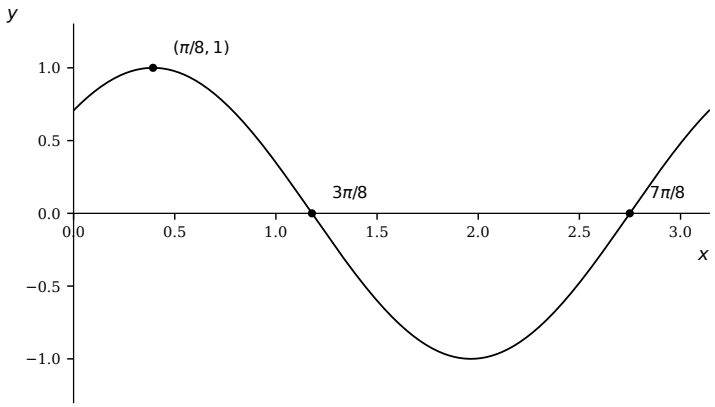
M1	输入范围
M1	零点
M1	还原x
A1	两解

(b) · 3 marks

Zeros as (a), maximum $(\pi/8, 1)$, minimum $(5\pi/8, -1)$; period π.

平移π/8、水平缩短为周期π，标出顶点及谷点，端点为$\sqrt{2}/2$。

M1	相位与周期
B1	极值点
A1	草图



Suggested sketch

Question 20

P3-C04-R20 · 6 marks

(a) · 3 marks

$\tan(\pi/4) = 1$ gives the stated expression.

正确引用加角公式，将第二角正切设1，并保留分母非零。

M1	正确和角公式
M1	精确值代入
A1	目标式

(b) · 3 marks

$x = \arctan(1/2)$.

(1+t)/(1−t)=3给t1/2；在锐域唯一且x+π/4不等于π/2。

M1	线性分式求tanx
M1	分支和域检查
A1	精确角

Question 21

P3-C04-R21 · 7 marks

(a) · 4 marks

Maximum 15; $x+\alpha=\pi/2+2k\pi$ with $\alpha=\arctan(4/3)$.

辅助角R15，Rcosα9、Rsinα12，正弦最高在π/2加整周。

M1	辅助角R
M1	正确相位
M1	最大输入
A1	值及方程

(b) · 3 marks

30.

加π使sin与cos都变号，P(x+π)=−P(x)，Q=2P，所以最高30。

M1	半周变号
M1	Q等于2P
A1	最大值

Question 22

P3-C04-R22 · 7 marks

(a) · 4 marks

$x = 0, \pi/6, \pi.$

sinx(2cosx−√3)=0，包含两端sin0，以及cos√3/2的π/6。

M1	双角分解
M1	零因子与端点
M1	余弦支
A1	三角

(b) · 3 marks

$x = \pi/6;\ \cos 2x = 1/2.$

去除0和π后唯一π/6，双倍输入π/3。

B1	内部角
M1	双角输入
A1	值

Question 23

P3-C04-R23 · 5 marks

(a) · 3 marks

7.

$\cos\theta=-24/25$，用tan半角=$\sin\theta/(1+\cos\theta)=(7/25)/(1/25)$。

M1	正确余弦负根
M1	半角关系
A1	比值

(b) · 2 marks

$1/50$.

$(1+\cos\theta)/2=(1/25)/2$。

M1	半角平方公式
A1	精确值

Question 24

P3-C04-R24 · 7 marks

(a) · 4 marks

$\sin 2x\cos x+\cos 2x\sin x=2\sin x\cos^2 x+(1-2\sin^2 x)\sin x=3\sin x-4\sin^3 x.$

依次加角、双角、平方和替换，完整显示推导，不直接把三倍角当已知。

M1	加角展开
M1	双角替换
M1	化为sin幂
A1	目标式

(b) · 3 marks

$x=0,\pi/4.$

用(a)化为$2\sin x(1-2\sin^2 x)=0$，正域内$\sin 0$或$\sqrt{2}/2$。

M1	引用(a)分解
M1	角范围筛选
A1	两角

Question 25

P3-C04-R25 · 6 marks

(a) · 4 marks

$\sin x = 0$; $x = 0, \pi$.

两式差为−2sinxsinπ/3=0，sinπ/3非零，整周期半开域两根。

M1	两余弦展开
M1	简化
M1	端点筛选
A1	两解

(b) · 2 marks

$1/2$ at 0; $-1/2$ at π.

代原相移余弦，输入π/3和4π/3。

M1	原式代入
A1	两值

Question 26

P3-C04-R26 · 8 marks

(a) · 3 marks

$R = 10$, $\alpha = \arctan(4/3)$.

比较Rcosα6和Rsinα8，减角公式提供负余弦项。

M1	正确对应
M1	R
A1	alpha

(b) Find

(i) · 3 marks

Minimum 0 at $t = 3\pi/2 + \alpha$ in the stated interval.

sin相位=−1时输入3π/2；加α<π/2仍小于2π。

M1	最低正弦
M1	选择域内相位
A1	高度和t

(ii) · 2 marks

π.

等价sin(t−α)>0，在完整周长2π内占π长度。

M1	基准线以上正弦条件
A1	总长度

Question 27

P3-C04-R27 · 6 marks

(a) · 3 marks

$(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) = \cos 2x.$

先提平方差，再用平方和1和双角公式。

M1	平方差
M1	两恒等式
A1	结果

(b) · 3 marks

$x = \pi/3, 2\pi/3.$

cos2x=−1/2，输入在0至2π的2π/3、4π/3，除2。

M1	使用(a)
M1	倍角范围筛选
A1	两解

Question 28

P3-C04-R28 · 6 marks

(a) · 3 marks

Sum condition $20(\sin\theta+\cos\theta)=28$; $\sin 2\theta = 24/25$.

对和平方得1+sin2θ=49/25，所以24/25。

M1	两投影关系
M1	平方恒等式
A1	双角值

(b) · 3 marks

$36.9^\circ, 53.1^\circ$.

两锐角互余，sinθ和cosθ为3/5、4/5两种配对；也可解双角0至180°。

M1	两分支与锐域
M1	反求角
A1	两角

Question 29

P3-C04-R29 · 7 marks

(a) · 3 marks

$R = 10,\ \alpha = \arctan(3/4) = 0.6435.$

展开$R\cos(x-\alpha) = R\cos\alpha\cos x + R\sin\alpha\sin x$，比较得$R\cos\alpha = 8, R\sin\alpha = 6$。

M1	正确展开比较系数
A1	R
A1	alpha

(b) The horizontal line $y = k$ meets the curve at two distinct points in $0 \le x < 2\pi$.

(i) · 2 marks

$-10 < k < 10.$

每个内部高度在完整周期内有两个交点；在极大或极小高度仅有一个切点，超出值域无交点。半开区间避免重复周期端点。

B1	正确范围
B1	排除两端理由

(ii) · 2 marks

$x = 1.691, 5.879.$

$\cos(x-\alpha) = 1/2$，所以$x = \alpha \pm \pi/3$并调整到$[0, 2\pi)$；两解为$1.69069\ldots, 5.87949\ldots$。

M1	完整余弦解分支
A1	两个横坐标

Question 30

P3-C04-R30 · 7 marks

(a) · 3 marks

$\beta = \arctan(3/4)$.

竖直投影分别为$4\sin\theta$和$3\sin(\pi/2-\theta) = 3\cos\theta$。比较$5\cos\beta = 4, 5\sin\beta = 3$。

M1	投影与余角
M1	辅助角比较
A1	beta

(b) Find

(i) · 2 marks

$H_{\max} = 5$ m at $\theta = \arctan(4/3)$.

当$\theta + \beta = \pi/2$时正弦为1，角度位于允许区间；$\theta = \pi/2 - \beta = \arctan(4/3)$。

M1	最大值条件且合法
A1	高度与角

(ii) · 2 marks

$\theta = 0.476, 1.378$.

$\sin(\theta+\beta) = 0.9$。结合$\theta \in (0, \pi/2)$，两支为$\arcsin 0.9 - \beta$及$\pi - \arcsin 0.9 - \beta$，分别$0.47627\ldots, 1.37832\ldots$。

M1	两个正弦分支及区间
A1	两角